

EA-0023 - AQELYN Engineering Archive

AQELYN - EA-0023 Engineering Archive

IS-023 - AQELYN Threat Exposure & Attack Surface Management Engine

****Archive ID:**** EA-0023

****Implementation Specification:**** IS-023

****Component:**** AQELYN Threat Exposure & Attack Surface Management Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0022

****Next Specification:**** IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0023 |

| Specification | IS-023 - AQELYN Threat Exposure & Attack Surface Management Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0023.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-023 complete; EA-0023 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine
EA-0015	IS-015	AQELYN Security Operations (SOC) Engine
EA-0016	IS-016	AQELYN Digital Forensics Engine
EA-0017	IS-017	AQELYN Threat Detection & Analytics Engine
EA-0018	IS-018	AQELYN Automated Response & Orchestration Engine
EA-0019	IS-019	AQELYN Security Data Lake & Telemetry Platform
EA-0020	IS-020	AQELYN AI Decision Intelligence Engine
EA-0021	IS-021	AQELYN Predictive Analytics & Forecasting Engine
EA-0022	IS-022	AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023	IS-023	AQELYN Threat Exposure & Attack Surface Management Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-023 Specification Identity

```
Specification ID: IS-023  
Name: AQELYN Threat Exposure & Attack Surface Management Engine  
Engineering Archive Target: EA-0023  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-022 - AQELYN Executive Intelligence & Strategic Reporting Engine
```

3. Purpose

The AQELYN Threat Exposure & Attack Surface Management Engine provides continuous discovery, inventory, analysis, prioritization, and governance of the organization's cyber attack surface. It identifies exposed assets, services, identities, applications, cloud resources, APIs, and external-facing infrastructure to reduce organizational exposure before exploitation.

The engine enables proactive exposure management by combining asset intelligence, threat intelligence, risk intelligence, predictive analytics, and executive governance.

It answers:

```
Which assets are externally exposed?  
What is our current attack surface?  
Which exposures present the highest business risk?  
Which assets require immediate remediation?  
How is the attack surface evolving?  
Which cloud resources are publicly accessible?  
Which identities increase exposure?  
Can every exposure be explained and audited?
```

4. Mission

The engine shall provide:

```
Attack surface discovery  
Exposure analysis  
External asset inventory  
Cloud exposure analysis  
Identity exposure analysis  
API exposure analysis  
Exposure prioritization  
Continuous monitoring  
Executive exposure summaries  
Exposure governance  
Exposure trend analysis  
Exposure auditing
```

5. Scope

5.1 In Scope

```
Internet-facing assets  
Cloud assets  
Applications  
Public APIs  
Domains  
Certificates  
Identity exposure  
Network exposure  
Infrastructure exposure  
Attack surface trends  
Exposure scoring
```

5.2 Out of Scope

```
Penetration testing  
Active exploitation  
Offensive operations  
Consumer asset inventory  
Non-security asset management
```

6. Dependencies

IS-023 depends on:

- IS-001 AQELYN Kernel
- IS-002 Universal Object Model
- IS-003 AQELYN Event Bus
- IS-004 AQELYN Evidence Engine
- IS-005 AQELYN Knowledge Graph
- IS-006 AQELYN Trust Engine
- IS-007 AQELYN Mission Engine
- IS-008 AQELYN Workflow Engine
- IS-009 AQELYN Policy Engine
- IS-010 AQELYN Compliance & Governance Engine
- IS-011 Identity & Access Governance Engine
- IS-012 Asset & Configuration Governance Engine
- IS-013 Risk Intelligence Engine
- IS-014 Threat Intelligence Fusion Engine
- IS-015 Security Operations Engine
- IS-016 Digital Forensics Engine
- IS-017 Threat Detection & Analytics Engine
- IS-018 Automated Response & Orchestration Engine
- IS-019 Security Data Lake & Telemetry Platform
- IS-020 AI Decision Intelligence Engine
- IS-021 Predictive Analytics & Forecasting Engine
- IS-022 Executive Intelligence & Strategic Reporting Engine

7. High-Level Architecture

AQELYN Threat Exposure & Attack Surface Management Engine

-
- Asset Discovery Engine
- Exposure Analysis Engine
- Attack Surface Engine
- Exposure Scoring Engine
- Cloud Exposure Engine
- Identity Exposure Engine
- API Exposure Engine
- Trend Analysis Engine
- Knowledge Graph Connector
- Data Lake Connector
- Risk Connector
- AI Decision Connector
- Event Publisher

8. Functional Requirements

FR-023-001 - Asset Discovery

The engine shall continuously discover:

- Internet-facing systems
- Cloud resources
- Applications
- Domains
- Certificates
- APIs
- Identity endpoints

FR-023-002 - Exposure Analysis

Analyze:

- Network exposure
- Identity exposure
- Application exposure

Cloud exposure
API exposure
Configuration exposure

FR-023-003 - Exposure Prioritization

Generate:

Exposure scores
Business impact
Mission impact
Risk ranking
Remediation priority
Executive exposure summaries

FR-023-004 - Explainable Exposure Intelligence

Every exposure shall include:

Evidence references
Confidence indicators
Risk rationale
Threat context
Asset lineage

FR-023-005 - Governance

Support:

Approval workflows
Version control
Auditability
Policy validation
Executive review

FR-023-006 - Event Publication

Publish standardized events:

asset.discovered
exposure.detected
attack_surface.updated
exposure.score.updated
cloud.exposure.detected
identity.exposure.detected

9. Non-Functional Requirements

The engine shall provide:

Continuous monitoring
Enterprise scalability
Low-latency discovery
Explainability
Auditability
Repository stability
Backward compatibility

10. Core Exposure Lifecycle

Asset Discovery
↓

```

Exposure Analysis
↓
Risk Scoring
↓
Threat Correlation
↓
Policy Validation
↓
Executive Review
↓
Continuous Monitoring

```

11. Internal Component Architecture

The AQELYN Threat Exposure & Attack Surface Management Engine is implemented as a modular exposure management platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Risk Intelligence Engine, AI Decision Intelligence Engine, Identity Governance Engine, Asset Governance Engine, and Threat Intelligence Fusion Engine.

AQELYN Threat Exposure & Attack Surface Management Engine

```

■
■■■ Asset Discovery Engine
■■■ Exposure Analysis Engine
■■■ Attack Surface Engine
■■■ Exposure Scoring Engine
■■■ Cloud Exposure Engine
■■■ Identity Exposure Engine
■■■ API Exposure Engine
■■■ Trend Analysis Engine
■■■ Knowledge Graph Connector
■■■ Data Lake Connector
■■■ Risk Connector
■■■ AI Decision Connector
■■■ Event Publisher

```

12. Component Specifications

12.1 Asset Discovery Engine

Continuously discovers organizational assets.

Capabilities:

```

Internet-facing assets
Cloud assets
Applications
Domains
Certificates
APIs

```

12.2 Exposure Analysis Engine

Analyzes organizational exposure.

Supports:

```

Network exposure
Application exposure
Identity exposure
Cloud exposure
Configuration exposure

```

12.3 Attack Surface Engine

Maintains the enterprise attack surface inventory.

Supports:

- Attack surface mapping
- Asset relationships
- Exposure inventory
- Surface evolution
- Discovery history

12.4 Exposure Scoring Engine

Calculates exposure risk.

Produces:

- Exposure score
- Business impact
- Mission impact
- Risk priority
- Remediation priority

12.5 Cloud Exposure Engine

Analyzes cloud exposure.

Supports:

- Public cloud assets
- Storage exposure
- Identity exposure
- Cloud networking
- Cloud services

12.6 Identity Exposure Engine

Evaluates identity-related exposure.

Produces:

- Privileged exposure
- Credential exposure
- External identity exposure
- Service account exposure
- Federation exposure

12.7 API Exposure Engine

Discovers exposed APIs.

Supports:

- Public APIs
- Shadow APIs
- Deprecated APIs
- API authentication analysis
- API inventory

12.8 Trend Analysis Engine

Analyzes attack surface and exposure changes over time.

Supports:

```
Exposure trend calculation
Attack surface growth analysis
Risk trend analysis
Cloud exposure trend analysis
Identity exposure trend analysis
```

13. Universal Object Model Extensions

13.1 ExposureRecord

```
ExposureRecord:
  exposure_id
  asset_id
  exposure_type
  risk_score
```

13.2 AttackSurfaceAsset

```
AttackSurfaceAsset:
  asset_id
  classification
  exposure_level
  discovered_at
```

13.3 ExposureTrend

```
ExposureTrend:
  trend_id
  category
  direction
  reporting_period
```

13.4 ExposureAssessment

```
ExposureAssessment:
  assessment_id
  confidence
  recommendations
  generated_at
```

14. Knowledge Graph Integration

Relationships:

```
Asset
↓
has
↓
Exposure
Exposure
↓
affects
↓
Mission
Mission
↓
impacts
↓
Risk
Risk
↓
guides
↓
Remediation
```

15. Event Bus Integration

15.1 Asset Events

```
asset.discovered  
asset.updated  
asset.removed
```

15.2 Exposure Events

```
exposure.detected  
exposure.updated  
exposure.closed
```

15.3 Attack Surface Events

```
attack_surface.updated  
attack_surface.analyzed
```

15.4 Cloud & Identity Events

```
cloud.exposure.detected  
identity.exposure.detected  
api.exposure.detected
```

16. Security Data Lake Integration

Consumes:

```
Asset telemetry  
Discovery history  
Configuration history  
Exposure history  
Network telemetry
```

17. AI Decision Intelligence Integration

Consumes:

```
Remediation recommendations  
Confidence scores  
Decision history  
Exposure prioritization  
Learning insights
```

18. Risk Intelligence Integration

Consumes:

```
Risk scores  
Business impact  
Mission impact  
Threat likelihood  
Risk trends
```

19. Compliance Integration

Supports:

- Exposure auditing
- Regulatory reporting
- Policy validation
- Governance reporting
- Traceability

20. Public APIs

20.1 Exposure API

- GET /exposures
- POST /exposures
- GET /exposures/{id}

20.2 Asset API

- GET /assets
- POST /assets

20.3 Attack Surface API

- GET /attack-surface
- POST /attack-surface/scan

20.4 Assessment API

- GET /assessments
- GET /exposure-summary

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/  
■■■ src/  
■ ■■■ attack_surface_management/  
■■■ tests/  
■ ■■■ attack_surface_management/  
■■■ docs/  
■ ■■■ attack_surface_management/  
■■■ api/  
■ ■■■ attack_surface_management/  
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Threat Exposure & Attack Surface Management Engine is the trusted subsystem responsible for continuously discovering, evaluating, prioritizing, and governing organizational cyber exposure across infrastructure, identities, applications, APIs, cloud resources, and external attack surfaces.

Every exposure record shall be:

- Explainable
- Evidence-backed
- Policy-governed
- Risk-aware
- Mission-aware
- Fully auditable
- Traceable
- Continuously validated

22.1 Security Principles

- Zero Trust
- Defense in Depth
- Least Privilege
- Continuous Discovery
- Explainable Exposure Intelligence
- Policy Enforcement
- Secure by Design
- Continuous Validation

22.2 Authorization Model

Supported operational roles:

- SOC Analyst
- Exposure Analyst
- Cloud Security Engineer
- Identity Administrator
- Risk Analyst
- Mission Owner
- Security Administrator
- Compliance Officer

All exposure information shall be governed through the AQELYN Policy Engine.

22.3 Exposure Integrity

Exposure records shall maintain:

- Unique exposure identifier
- Asset identifier
- Exposure type
- Risk score
- Evidence references
- Confidence indicators
- Discovery source
- Validation state
- Audit trail

Exposure history shall be append-only.

22.4 Attack Surface Integrity

Attack surface records shall maintain:

- Asset ownership
- Classification
- Exposure level
- Discovery timestamp
- Last validation timestamp
- Relationship context
- Threat context
- Policy references

No attack surface record shall be considered authoritative without source lineage and validation metadata.

23. Exposure Lifecycle

23.1 Discovery Lifecycle

```
graph TD; A[Asset Discovery] --> B[Exposure Detection]; B --> C[Risk Analysis]; C --> D[Exposure Classification]; D --> E[Policy Validation]; E --> F[Continuous Monitoring];
```

23.2 Remediation Lifecycle

```
graph TD; A[Exposure Prioritized] --> B[Recommendation Generated]; B --> C[Remediation Initiated]; C --> D[Exposure Revalidated]; D --> E[Exposure Closed];
```

23.3 Audit Lifecycle

```
graph TD; A[Exposure Detected] --> B[Evidence Linked]; B --> C[Policy Verified]; C --> D[Audit Stored];
```

24. Continuous Exposure Operations

The engine continuously evaluates:

```
graph TD; A[Internet-facing assets] --> B[Cloud exposure]; B --> C[Identity exposure]; C --> D[API exposure]; D --> E[Network exposure]; E --> F[Exposure trends]; F --> G[Risk evolution]; G --> H[Policy compliance];
```

25. Performance Requirements

The engine shall support:

```
graph TD; A[Continuous asset discovery] --> B[Low-latency exposure analysis]; B --> C[Enterprise-scale inventory]; C --> D[Concurrent exposure assessments]; D --> E[Continuous operation]; E --> F[High availability];
```

26. Scalability Requirements

The engine shall scale to support:

- Global enterprises
- Multi-cloud deployments
- Hybrid infrastructure
- Large attack surfaces
- Millions of assets
- Long-term exposure history

27. Audit Requirements

Every exposure operation shall generate immutable audit records.

Audit events include:

- Asset discovered
- Exposure detected
- Exposure updated
- Exposure closed
- Risk recalculated
- Policy validation

28. Failure Handling

28.1 Discovery Failure

- Discovery retried
- Failure recorded
- Administrator notified

28.2 Exposure Analysis Failure

- Exposure analysis retried
- Fallback assessment generated
- Audit recorded

28.3 Risk Scoring Failure

- Risk recalculated
- Previous score retained
- Audit generated

28.4 Policy Failure

- Exposure publication blocked
- Policy violation recorded
- Human review required

29. Testing Strategy

29.1 Unit Testing

Validate:

```
Asset Discovery Engine
Exposure Analysis Engine
Attack Surface Engine
Exposure Scoring Engine
Cloud Exposure Engine
Identity Exposure Engine
API Exposure Engine
```

29.2 Integration Testing

Verify interaction with:

```
Kernel
Knowledge Graph
Security Data Lake
Risk Intelligence Engine
AI Decision Engine
Identity Governance Engine
Asset Governance Engine
Threat Intelligence Engine
Compliance Engine
```

29.3 System Testing

Validate:

```
Asset discovery
Exposure analysis
Attack surface inventory
Exposure scoring
Executive exposure summaries
Exposure auditing
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Exposure integrity
```

29.5 Regression Testing

Verify IS-001 through IS-022 remain unaffected.

30. Acceptance Criteria

IS-023 is complete when:

```
Asset Discovery Engine implemented
Exposure Analysis Engine implemented
Attack Surface Engine implemented
Exposure Scoring Engine implemented
Cloud Exposure Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/  
■■■ src/attack_surface_management/  
■■■ tests/attack_surface_management/  
■■■ docs/attack_surface_management/  
■■■ api/attack_surface_management/  
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-023 introduces the AQELYN Threat Exposure & Attack Surface Management Engine, providing enterprise-scale attack surface discovery, exposure analysis, continuous monitoring, cloud and identity exposure analysis, exposure scoring, explainable exposure intelligence, and policy-governed exposure management.

Major capabilities include:

```
Attack Surface Discovery  
Exposure Analysis  
Exposure Scoring  
Cloud Exposure Analysis  
Identity Exposure Analysis  
API Exposure Analysis  
Exposure Trends  
Risk Prioritization  
Executive Exposure Reporting  
Explainable Exposure Intelligence  
Continuous Monitoring
```

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

```
Specification ID : IS-023  
Title : AQELYN Threat Exposure & Attack Surface Management Engine  
Status : COMPLETE  
Engineering Archive : READY FOR GENERATION  
Next Artifact : EA-0023
```

Engineering workflow status:

```
EA-0001 COMPLETE  
EA-0002 COMPLETE  
EA-0003 COMPLETE  
EA-0004 COMPLETE  
EA-0005 COMPLETE  
EA-0006 COMPLETE  
EA-0007 COMPLETE  
EA-0008 COMPLETE  
EA-0009 COMPLETE  
EA-0010 COMPLETE  
EA-0011 COMPLETE  
EA-0012 COMPLETE  
EA-0013 COMPLETE  
EA-0014 COMPLETE  
EA-0015 COMPLETE  
EA-0016 COMPLETE  
EA-0017 COMPLETE  
EA-0018 COMPLETE  
EA-0019 COMPLETE  
EA-0020 COMPLETE  
EA-0021 COMPLETE  
EA-0022 COMPLETE  
IS-023 COMPLETE  
EA-0023 READY FOR GENERATION
```

34. EA-0023 Engineering Objective

The objective of IS-023 was to introduce a dedicated Threat Exposure & Attack Surface Management Engine that enables AQELYN to continuously discover, assess, prioritize, monitor, and govern external and internal cyber exposure.

The engine extends AQELYN from executive intelligence into proactive exposure reduction and attack surface governance.

35. EA-0023 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Asset discovery
- Exposure analysis
- Attack surface inventory
- Exposure scoring
- Cloud exposure analysis
- Identity exposure analysis
- API exposure analysis
- Trend analysis
- Knowledge Graph integration
- Security Data Lake integration
- Risk Intelligence integration
- AI Decision integration
- Compliance integration
- Event publishing

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Exposure Management Engine

Attack surface and exposure management responsibilities are implemented as a standalone engine rather than embedded in Asset Governance, Risk Intelligence, or Threat Detection.

Rationale:

- Clear separation of exposure discovery from asset inventory.
- Independent lifecycle and governance.
- Better support for continuous attack surface analysis.
- Improved prioritization of exposed assets before exploitation.

36.2 Decision 2 - Exposure Records Are Evidence-Backed

Every exposure record must reference source evidence, confidence indicators, risk rationale, and asset lineage.

Benefits:

- Exposure findings become auditable.
- Remediation prioritization becomes defensible.
- Compliance and executive reporting can consume exposure intelligence.

36.3 Decision 3 - Attack Surface Assets as First-Class Objects

Attack surface assets are modeled as Universal Object Model extensions.

Benefits:

External-facing assets become queryable.
 Exposure history can be tracked across time.
 Knowledge Graph can relate assets, missions, risks, and remediation.

36.4 Decision 4 - Event-Driven Exposure Lifecycle

Asset discovery, exposure detection, attack surface analysis, scoring, and closure events are published through the AQELYN Event Bus.

Examples include:

```
asset.discovered
asset.updated
asset.removed
exposure.detected
exposure.updated
exposure.closed
attack_surface.updated
attack_surface.analyzed
cloud.exposure.detected
identity.exposure.detected
api.exposure.detected
```

This maintains loose coupling between AQELYN engines.

36.5 Decision 5 - Universal Object Model Extension**New domain objects introduced include:**

```
ExposureRecord
AttackSurfaceAsset
ExposureTrend
ExposureAssessment
```

These extend the Universal Object Model without modifying existing object definitions.

37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Exposure, attack surface asset, trend, assessment objects |

| IS-003 Event Bus | Asset, exposure, attack surface, cloud, identity, API events |

| IS-004 Evidence Engine | Evidence references and exposure support |

| IS-005 Knowledge Graph | Asset, exposure, mission, risk, remediation relationships |

| IS-006 Trust Engine | Data confidence and exposure trust |

| IS-007 Mission Engine | Mission-aware exposure prioritization |

| IS-008 Workflow Engine | Exposure remediation and review workflows |

| IS-009 Policy Engine | Exposure governance, publication, and validation policies |

| IS-010 Compliance Engine | Exposure auditing and regulatory reporting |

| IS-011 Identity Governance Engine | Identity exposure and privileged exposure context |

| IS-012 Asset Governance Engine | Asset ownership, criticality, classification |

| IS-013 Risk Intelligence Engine | Risk scoring and business impact |

| IS-014 Threat Intelligence Engine | Threat likelihood and attack context |

IS-019 Security Data Lake	Discovery history, telemetry, exposure history
IS-020 AI Decision Engine	Remediation recommendations and learning insights
IS-022 Executive Reporting Engine	Executive exposure summaries

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/attack_surface_management/  
■■■ tests/attack_surface_management/  
■■■ api/attack_surface_management/  
■■■ docs/attack_surface_management/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces exposure-management-specific security controls:

```
Policy-governed exposure visibility  
Evidence-backed exposure records  
Continuous exposure validation  
Exposure scoring and prioritization  
Source lineage for discovery  
Attack surface inventory integrity  
Exposure remediation audit trail  
Cloud, identity, and API exposure tracking  
Role-authorized exposure administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Attack surface discovery  
Exposure analysis  
External asset inventory  
Cloud exposure analysis  
Identity exposure analysis  
API exposure analysis  
Exposure prioritization  
Continuous monitoring  
Executive exposure summaries  
Exposure governance  
Exposure trend analysis  
Exposure auditing
```

41. Risks Identified

| Risk | Mitigation |

|---|---|

Discovery gaps	Continuous discovery and multiple telemetry sources
False exposure findings	Confidence indicators and validation lifecycle
Excessive alerting	Exposure scoring and prioritization
Unauthorized exposure visibility	Policy enforcement and role authorization
Stale attack surface inventory	Continuous monitoring and validation timestamps
Poor remediation prioritization	Mission, risk, and threat context
Cloud misclassification	Cloud exposure engine and source lineage
API blind spots	API exposure engine and shadow API discovery

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover asset discovery, exposure analysis, attack surface engine, exposure scoring, cloud exposure engine, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

44. Dependencies

Required:

EA-0001 through EA-0022
IS-001 through IS-022

Enables:

IS-024 and subsequent vulnerability intelligence, prioritization, asset discovery, cloud posture, identity exposure, and attack surface governance components

45. Completion Record

```

Engineering Archive : EA-0023
Implementation Specification : IS-023
Title : AQELYN Threat Exposure & Attack Surface Management Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

```

46. Archive Index Update

```

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
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EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
EA-0022 IS-022 AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023 IS-023 AQELYN Threat Exposure & Attack Surface Management Engine

```

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0023

Current Status:
EA-0023 COMPLETE

Next Implementation Specification:
IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine

EA-0023 is completed and archived. The engineering workflow is consistent with the project rule:

```
Implementation Specification -> Engineering Archive -> Continue
```

From this point onward, the next engineering artifact is IS-024.

48. Engineering Archive Publication Standard

EA-0023 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```

EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg

```

```
■ ■ ■ ■ Component.svg
■ ■ ■ ■ Workflow.svg
■ ■ ■ ■ EventFlow.svg
■ ■ ■ ■ Integration.svg
■
■ ■ ■ ■ examples/
■ ■ ■ ■ example_*.md
■
■ ■ ■ ■ HTML/
■ ■ ■ ■ index.html
■ ■ ■ ■ styles.css
■ ■ ■ ■ assets/
■
■ ■ ■ ■ images/
■
■ ■ ■ ■ journal/
■ ■ ■ ■ Engineering_Journal.md
■
■ ■ ■ ■ manifest/
■ ■ ■ ■ manifest.json
■
■ ■ ■ ■ MD/
■ ■ ■ ■ EA-xxxx.md
■
■ ■ ■ ■ PDF/
■ ■ ■ ■ EA-xxxx.pdf
■
■ ■ ■ ■ requirements/
■ ■ ■ ■ Requirements_Matrix.md
■
■ ■ ■ ■ traceability/
■ ■ ■ ■ Traceability_Matrix.md
■
■ ■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID Requirement Evidence in Archive Status			
--- --- --- ---			
FR-023-001	Continuously discover assets	Sections 8, 12, 23	Complete
FR-023-002	Analyze exposure	Sections 8, 12, 23	Complete
FR-023-003	Prioritize exposure	Sections 8, 12, 18	Complete
FR-023-004	Provide explainable exposure intelligence	Sections 8, 22, 36	Complete
FR-023-005	Support governance	Sections 8, 22, 23	Complete
FR-023-006	Publish exposure events	Sections 8, 15, 36	Complete
NFR-023-001	Continuous monitoring	Sections 9, 24	Complete
NFR-023-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-023-003	Low-latency discovery	Sections 9, 25	Complete
NFR-023-004	Explainability	Sections 9, 22, 36	Complete
NFR-023-005	Auditability	Sections 9, 27, 39	Complete
NFR-023-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source Target Relationship		
--- --- ---		

| IS-023 Purpose | EA-0023 Objective | Defines why the engine exists |

| Asset Discovery Engine | FR-023-001 | Discovers attack surface assets |

| Exposure Analysis Engine | FR-023-002 | Analyzes exposure |

| Attack Surface Engine | FR-023-001, FR-023-002 | Maintains attack surface inventory |

| Exposure Scoring Engine | FR-023-003 | Calculates exposure priority |

| Cloud Exposure Engine | FR-023-002 | Analyzes cloud exposure |

| Identity Exposure Engine | FR-023-002 | Analyzes identity exposure |

| API Exposure Engine | FR-023-002 | Analyzes API exposure |

| Event Publisher | FR-023-006 | Publishes exposure events |

| Security Data Lake Integration | Discovery history | Supplies telemetry and exposure history |

| AI Decision Integration | Remediation recommendations | Supplies recommendations and confidence |

| Risk Intelligence Integration | Risk-aware prioritization | Supplies risk and mission impact |

| Compliance Integration | Exposure audit | Supports regulatory and policy reporting |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0023

EA-0023 was created to archive completion of IS-023 - AQELYN Threat Exposure & Attack Surface Management Engine.

The archive records the expansion of AQELYN into threat exposure and attack surface management. IS-023 defines the structure needed to discover internet-facing assets, analyze network, cloud, identity, application, configuration, and API exposure, maintain an attack surface inventory, score exposure risk, prioritize remediation, support executive exposure reporting, and publish exposure events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Exposure management must be modeled separately from asset inventory and vulnerability management. Asset inventory identifies what exists, vulnerability intelligence identifies weakness, and exposure management identifies what is reachable, visible, risky, and likely to be targeted.

Governance Note

EA-0023 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Exposure Record

```
exposure_id: EXP-0001
asset_id: ASSET-1001
exposure_type: public_api
risk_score: 87
confidence: 0.91
evidence:
- evidence://api-discovery-1001
```

```
- evidence://network-telemetry-2001
```

52.2 Example Attack Surface Asset

```
asset_id: ASA-2001
classification: internet_facing
exposure_level: high
discovered_at: 2026-07-07T12:00:00Z
owner: cloud_security_team
```

52.3 Example Exposure Assessment

```
assessment_id: EXA-3001
confidence: 0.88
recommendations:
- restrict public API access
- validate authentication policy
- review certificate and domain ownership
generated_at: 2026-07-07T12:15:00Z
```

52.4 Example Exposure Event

```
{
  "event_type": "exposure.detected",
  "exposure_id": "EXP-0001",
  "asset_id": "ASSET-1001",
  "risk_score": 87,
  "source_engine": "aqelyn_threat_exposure_attack_surface_management_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0023.md
PDF/EA-0023.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_attack_surface.md
```

54. Final Archive Statement

EA-0023 is the Engineering Archive for IS-023 - AQELYN Threat Exposure & Attack Surface Management Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0023 COMPLETE
IS-023 COMPLETE
NEXT: IS-024
```